

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CE/IT) Examination

May 2014

IT205 Operating System

Date: 07.05.2014, Wednesday

Time: 10:00 a.m. To 01: 00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose one of the alternatives for the following questions:

[05]

- (a) The root directory of a disk should be placed
 - a) At a fixed address in main memory
 - b) At a fixed location on the disk
 - c) Anywhere on the disk
 - d) At a fixed location on the system disk
 - e) Anywhere on the system disk
- (b) Which scheduling policy is most suitable for a time shared operating system?
 - a) Shortest Job First
 - b) Round Robin
 - c) First Come First Serve
 - d) Elevator
- (c) A critical section is a program segment
 - a) which should run in a certain specified amount of time.
 - b) which avoids deadlock
 - c) where shared resources are accessed
 - d) which must be enclosed by a pair of semaphore operations, P and V
- (d) When an interrupt occurs, an operating system
 - a) ignores the interrupt
 - b) always change the state of interrupted process after processing the interrupt
 - c) always resume execution of interrupted process after processing the interrupt
 - d) may change state of interrupted process to 'block' and schedule another process
- (e) When the result of a computation depends on the speed of the processes involved there is said to be
 - a) cycle stealing
 - b) race condition
 - c) a time lock
 - d) a deadlock

Q - 2 (a) State and explain process scheduling criteria.

[05]

(b) Explain Readers and Writers problem with solution.

[05]

(c) Consider the following set of processes A, B, C and D with the length of CPU Burst time given in milliseconds. Draw Gantt Chart for following Scheduling algorithms and calculate average waiting time and turn-around time:

[05]

- 1) Round Robin (Time Quantum = 4)
- 2) Shortest Job First (Pre-emptive).

Process	Arrival Time	Burst Time
A	0	8
B	1	4
C	2	9
D	3	5

OR

- (b) Explain Producer-Consumer Problem and its solution using semaphore. [05]
- (c) Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared boolean variable S1 and S2 are randomly assigned. [05]

Method used by P1	Method used by P2
While (S1 == S2); Critical Section S1 = S2;	While (S1 != S2); Critical Section S2 = not(S1);

Which requirements of the critical section are achieved? Justify your answer.

Q - 3 Attempt Any Five:

[15]

- What is thread? Differentiate between user-level thread and kernel-level thread.
- Draw the memory hierarchy with respect to speed, capacity and cost per bit.
- List out the types of operating system.
- State and explain file allocation methods.
- What is i-node? Explain i-node structure in UNIX.
- Differentiate: deadlock and starvation with example.
- Differentiate: Semaphore and Monitor.

SECTION – II

Q - 4 Do as Directed:

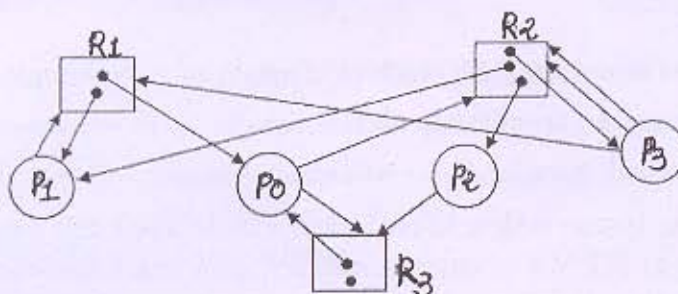
[05]

- Which page replacement policy sometimes leads to more page faults when size of memory is increased?
- Which of the following is/are advantage of virtual memory?
 - faster access to memory on an average
 - processes can be given protected address space
 - linker can assign addresses independent of where the program will be loaded in physical memory
 - programs larger than the physical memory size can be run
- Increasing the RAM of a computer typically improves the performance because:
 - Virtual Memory increases
 - Larger RAMs are faster
 - Fewer page fault occurs
 - Fewer segmentation fault occurs

- (d) An operating system implements a policy that requires a process to release all resources before making a request for another resource. Select TRUE statement from the following:
- Both starvation and deadlock can occur
 - Starvation can occur but deadlock cannot occur
 - Starvation cannot occur but deadlock can occur
 - neither starvation nor deadlock can occur
- (e) RAID level ____ spreads parity and data among all $N+1$ disks rather than storing data in N disks and parity in 1.

Q - 5 (a) Consider the resource allocation graph given in the figure.

[07]



- Find whether the system is in deadlock state or not.
 - If not then find the safe sequence.
- (b) The head of a moving head disk with 100 tracks numbered from 0 to 99 is currently serving a request at track 55. If the queue of requests kept in FIFO order is

10, 70, 75, 23, 65

Which of the two disk scheduling algorithms FCFS and SSTF will require less lead movement? Find the head movement for each of the algorithm.

- (c) A computer installation has 1000k of main memory. The jobs arrive and finish in the following sequences.

Job 1 requiring 200k arrives
 Job 2 requiring 350k arrives
 Job 3 requiring 300k arrives
 Job 1 finishes
 Job 4 requiring 120k arrives
 Job 5 requiring 150k arrives
 Job 6 requiring 80k arrives

- Draw the memory allocation table using best fit and first fit algorithms.
- Which algorithm performs better for this sequence?

OR

- (b) A process references five pages A, B, C, D and E in the following order:

[04]

A; B; C; D; A; B; E; A; A; B; C; D; E

Find the number of page fault during this sequence of references starting with an empty main memory with three page frames using First-In-First-Out and Least Recently Used algorithms.

(c) Consider the following segment table:

[04]

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

- 1) 0, 430 2) 1, 10 3) 2, 500 4) 4, 112

Q – 6 **Attempt Any Five:**

[15]

- (a) What are the ways of recovering from deadlock? Explain all with example.
- (b) Explain Direct Memory Access in detail.
- (c) Explain block device and character device with examples.
- (d) Consider a computer system with a 32-bit logical address and 4-KB page size. The system supports up to 512 MB of physical memory. How many entries are there in each of the following?
 - a) A conventional single level page table
 - b) An inverted page table
- (e) How do we select page size? Derive an equation to calculate page size.
- (f) Explain Resource Allocation Graph.
- (g) Compare local and global allocation policies for page replacement algorithms.
